Walchand College of Engineering, Sangli

Computer Science & Engineering

Third Year

Course: Computer Algorithm Lab

Assignment No. 6

Dynamic Programming

1. A warehouse robot starts at the top-left corner of a grid (m × n) and must reach the bottom-right corner to pick up a package. The robot can only move **right** or **down** at each step. Some cells contain obstacles (broken tiles) the robot cannot pass through.

   Task: Find the number of distinct paths the robot can take to reach the package.

   Input: Grid of size m × n with 0 (free) and 1 (obstacle). **Output:** Number of valid paths.

2. You're packing a backpack for a trek with a weight capacity of W kg. You have a list of n items, each with a weight and a "usefulness value." You can either take an item completely or leave it — no partial items.

   Task: Maximize total usefulness value without exceeding weight capacity W.

   Input: n items with (weight, value) pairs, capacity W. **Output:** Maximum achievable value.

3. A currency kiosk at an airport wants to give change using the minimum number of currency notes/coins from a given set of denominations available in unlimited supply.

   Task (Part A): Find the minimum number of coins needed to make a given amount. If not possible, return -1. **Task (Part B):** Find the total number of distinct ways to make that amount.

   Input: List of denominations, target amount. **Output:** Minimum coins (Part A) and number of combinations (Part B).

4. A university uses a plagiarism detection tool that compares two submitted essays (as sequences of words or characters) to find how similar they are, based on the longest sequence of words appearing in the same relative order in both documents.

   Task: Find the length of the Longest Common Subsequence between the two essays, and reconstruct the subsequence itself.

   Input: Two strings/sequences A and B. **Output:** Length of LCS + the actual subsequence.

5. A biotech lab is comparing a mutated DNA sequence to a reference sequence. They want to know the minimum number of operations (insert, delete, substitute a base) required to convert the mutated sequence into the reference sequence.

   Task: Compute the minimum edit distance between two strings.

   Input: Two strings representing DNA sequences. **Output:** Minimum number of edit operations.